

Frederick Stock

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Experience

University of Massachusetts Lowell

Fall 2024 – Current

Research Assistant

- Conducting research on reconfiguration of modular robots under Hugo Akitaya, funded by NSF grant CCF-2348067.

University of Massachusetts Lowell

Fall 2023 – Spring 2024

Teaching Assistant

- Teaching assistant for two sections of “COMP 5030: Algorithms” (graduate level) and one section of “COMP 4030: Algorithms”. Duties involved hosting weekly office hours, grading homeworks, and running weekly recitations.

Education

University of Massachusetts Lowell

2022 – 2026

Ph.D. in Computer Science

Rochester Institute of Technology

2017 – 2021

Bachelor of Science in Applied Mathematics

References

Hugo Akitaya

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Assistant Professor

University of Massachusetts Lowell

Joseph O’Rourke

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Professor (Emeritus)

Smith College

Amanda Redlich

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Assistant Professor

University of Massachusetts Lowell

Csaba Tóth

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Professor

CSUN Northridge & Tufts University

Service

Open Problem Transcription: The Canadian Conference on Computational Geometry (CCCG) 2025, 2024

Technical and Administrative Support: Fall Workshop on Computational Geometry (FWCG) 2024

Reviewer: Innovations in Theoretical Computer Science (ITCS) 2026

Reviewer: International Conference and Workshops on Algorithms and Computation (WALCOM) 2026

Reviewer: International Symposium on Distributed Computing (DISC) 2025

Reviewer: International Symposium on Computational Geometry (SoCG) 2025, 2024

Reviewer: Journal on Computational Geometry (JoCG) 2025

Reviewer: The Fall Workshop on Computational Geometry (FWCG) 2024

Reviewer: The Canadian Conference on Computational Geometry (CCCG) 2024

Reviewer: International Symposium on Algorithms and Computation (ISAAC) 2024

Reviewer: Journal of Information Processing (JCDCGGG Special Issue) (JIP) 2024

Publications

Upcoming Publications

1. “Redistricting in Triangular and Square Grids” (Hugo Akitaya, Kyle Dituto, Andrei Gonczi, Matias Korman, Diane Souvaine, Frederick Stock & Csaba D. Tóth), *Symposium on Simplicity in Algorithms (SOSA)*, 2026
2. “Super Guarding and Dark Rays in Art Galleries” (MIT CompGeom Group, Hugo Akitaya, Erik Demaine, Adam Hesterberg, Anna Lubiw, Jayson Lynch, Joseph O’Rourke & Frederick Stock), *Discrete Mathematics & Theoretical Computer Science (DMTCS)*, TBA
3. “A Universal In-Place Reconfiguration Algorithm for Sliding Cube-Shaped Robots in a Quadratic Number of Moves” (Zachary Abel, Hugo Akitaya, Scott Duke Kominers, Matias Korman & Frederick Stock), *The Journal of Computational Geometry (JoCG)*, TBA
4. “Agent Motion Planning as Block Asynchronous Cellular Automata: Pushing, Pulling, Suplexing, and More” (MIT Hardness Group, Hayashi Layers, Josh Brunner, Erik Demaine, Jenny Diomidova, Timothy Gomez, Della Hendrickson, Yael Kirkpatrick, Jeffery Li, Jayson Lynch, Ritam Nag & Frederick Stock), *Natural Computing (NACO)*, TBA

Journal Papers

5. “Minimum Plane Bichromatic Spanning Trees” (Hugo Akitaya, Ahmad Biniaz, Erik Demaine, Linda Kleist, Frederick Stock & Csaba D. Tóth), *ACM Transactions on Algorithms (Transactions on Algorithms)*, 2024
6. “Physical Visitor Access Control and Authentication Using Blockchain, Smart Contracts and Internet of Things” (Frederick Stock, Yesem Kurt Peker, Alfredo J. Perez & Jarel Hearst), *Cryptography (MDPI), Special Issue on Emerging Topics in Blockchain Security and Privacy (Cryptography)*, 2022

Conference Proceedings

7. “Connectivity Augmentation for Planar Graphs and Beyond-Planar Graphs” (Hugo Akitaya, Justin Dallant, Erik Demaine, Michael Kaufmann, Linda Kleist, Frederick Stock, Csaba D. Tóth & Torsten Ueckerdt), *The 33rd International Symposium on Graph Drawing and Network Visualization (GD)*, 2025
8. “Sliding Squares in Parallel” (Hugo Akitaya, Sándor Fekete, Peter Kramer, Saba Molaei, Christian Rieck, Frederick Stock & Tobias Wallner), *The European Symposium on Algorithms (ESA)*, 2025
9. “Input-Sensitive Reconfiguration of Sliding Cubes” (Hugo Akitaya, Matias Korman & Frederick Stock), *The Canadian Conference on Computational Geometry (CCCG)*, 2025
10. “Quasigeodesics on the Cube” (MIT CompGeom Group, Hugo Akitaya, Erik Demaine, Adam Hesterberg, Thomas C. Hull, Anna Lubiw, Jayson Lynch, Klara Mundilova, Chie Nara, Joseph O’Rourke, Frederick Stock, Josef Tkadlec & Ryuhei Uehara), *The Canadian Conference on Computational Geometry (CCCG)*, 2025
11. “On Inside-out Dissections of Polygons and Polyhedra” (Reymond Akpanya, Adi Rivkin & Frederick Stock), *The Canadian Conference on Computational Geometry (CCCG)*, 2025
12. “Brief Announcement: Broadcast via Mobile Agents in a Dynamic Network: Interplay of Graph Properties & Agents” (William K. Moses Jr., Amanda Redlich & Frederick Stock), *4th Symposium on Algorithmic Foundations of Dynamic Networks (SAND)*, 2025
13. “Finding Shortest Reconfiguration Sequences for Modular Robots” (UML Modular Robotics Group, Hugo Akitaya, Andrew Clements, Sam Downey, Jonathan Eisenbies, Soham Samanta, Gabriel Shahrouzi & Frederick Stock), *41st International Symposium on Computational Geometry Media Exposition (SoCG:ME 2025)*, 2025
14. “Navigation in dynamic graphs: How to defeat the Acme Graph Exploder” (Amanda Redlich, William K. Moses Jr. & Frederick Stock), *2025 Spring Eastern Sectional Meeting of the American Mathematical Society (AMS Spring 2025)*, 2025

15. “Minimum Plane Bichromatic Spanning Trees” (Hugo Akitaya, Ahmad Biniaz, Erik Demaine, Linda Kleist, Frederick Stock & Csaba D. Tóth), *35th International Symposium on Algorithms and Computation (ISAAC)*, 2024
16. “Easier Ways to Prove Counting Hard: A Dichotomy for Generalized #SAT, Applied to Constraint Graphs” (MIT Hardness Group, Josh Brunner, Erik Demaine, Jenny Diomidova, Timothy Gomez, Markus Hecher, Frederick Stock & Zixiang Zhou), *35th International Symposium on Algorithms and Computation (ISAAC)*, 2024
17. “Agent Motion Planning as Block Asynchronous Cellular Automata: Pushing, Pulling, Suplexing, and More” (MIT Hardness Group, Hayashi Layers, Josh Brunner, Erik Demaine, Jenny Diomidova, Timothy Gomez, Della Hendrickson, Yael Kirkpatrick, Jeffery Li, Jayson Lynch, Ritam Nag & Frederick Stock), *21st International Conference on Unconventional Computation and Natural Computation (UCNC)*, 2024
18. “A Universal In-Place Reconfiguration Algorithm for Sliding Cube-Shaped Robots in a Quadratic Number of Moves” (Zachary Abel, Hugo Akitaya, Scott Duke Kominers, Matias Korman & Frederick Stock), *40th International Symposium on Computational Geometry (SoCG)*, 2024
19. “Deltahedral Domes over Equiangular Polygons” (MIT CompGeom Group, Hugo Akitaya, Erik Demaine, Adam Hesterberg, Anna Lubiw, Jayson Lynch, Joseph O’Rourke, Frederick Stock & Josef Tkadlec), *40th International Symposium on Computational Geometry (EuroCG)*, 2024
20. “Super Guarding and Dark Rays in Art Galleries” (MIT CompGeom Group, Hugo Akitaya, Erik Demaine, Adam Hesterberg, Anna Lubiw, Jayson Lynch, Joseph O’Rourke & Frederick Stock), *The Canadian Conference on Computational Geometry (CCCG)*, 2023
21. “Blockchain Ensured Physical Visitor Access Control and Authentication” (Frederick Stock, Jarel Hearst & Yesem Kurt Peker), *IEEE International Conference on Mobile Ad Hoc and Smart Systems, REUNS (IEEE MASS REUNS)*, 2022

Other Publications

22. “Open Problems from CCCG 2025” (Hugo Akitaya, Shahin Kamali, Seyed-Mohammad Seyed-Javadi & Frederick Stock), *The Canadian Conference on Computational Geometry (CCCG)*, 2025
23. “Hardness Table Layout Hardness Table” (MIT Hardness Group, Josh Brunner, Erik Demaine, Jenny Diomidova, Della Hendrickson, Timothy Gomez, Hayashi Layers, Frederick Stock & Andy Tockman), *SIGTBD*, 2025
24. “Open Problems from CCCG 2024” (Reymond Akpanya, Bastien Rivier & Frederick Stock), *The Canadian Conference on Computational Geometry (CCCG)*, 2024
25. “Reconfiguration of 3D Pivoting Modular Robots” (Hugo Akitaya & Frederick Stock), *The International Symposium on Computational Geometry: Young Researchers Forum (SoCG:YRF)*, 2023

Collaborators

In the above have 52 collaborators: Zachary Abel, Hugo Akitaya, Reymond Akpanya, Ahmad Biniaz, Josh Brunner, Andrew Clements, Justin Dallant, Erik Demaine, Jenny Diomidova, Kyle Dituro, Sam Downey, Jonathan Eisenbies, Sándor Fekete, Timothy Gomez, Andrei Gonczi, Jarel Hearst, Markus Hecher, Della Hendrickson, Adam Hesterberg, Thomas C. Hull, William K. Moses Jr., Michael Kaufmann, Yael Kirkpatrick, Linda Kleist, Scott Duke Kominers, Matias Korman, Peter Kramer, Hayashi Layers, Jeffery Li, Anna Lubiw, Jayson Lynch, Saba Molaei, Klara Mundilova, Ritam Nag, Chie Nara, Joseph O’Rourke, Yesem Kurt Peker, Alfredo J. Perez, Amanda Redlich, Christian Rieck, Bastien Rivier, Adi Rivkin, Soham Samanta, Gabriel Shahrouzi, Diane Souvaine, Josef Tkadlec, Andy Tockman, Csaba D. Tóth, Torsten Ueckerdt, Ryuhei Uehara, Tobias Wallner, Zixiang Zhou

Presentations

1. “Input-Sensitive Reconfiguration of Sliding Cubes” *The Canadian Conference on Computational Geometry*, 2025
2. “On Inside-out Dissections of Polygons and Polyhedra” *The Canadian Conference on Computational Geometry*, 2025
3. “Finding Shortest Reconfiguration Sequences for Modular Robots” *41st International Symposium on Computational Geometry Media Exposition*, 2025
4. “Minimum Plane Bichromatic Spanning Trees” *35th International Symposium on Algorithms and Computation*, 2024
5. “Easier Ways to Prove Counting Hard: A Dichotomy for Generalized #SAT, Applied to Constraint Graphs” *35th International Symposium on Algorithms and Computation*, 2024
6. “Input-Sensitive Reconfiguration of Sliding Cubes” *The Fall Workshop on Computational Geometry*, 2024
7. “On Inside-out Dissections of Polygons and Polyhedra” *The Fall Workshop on Computational Geometry*, 2024
8. “A Universal In-Place Reconfiguration Algorithm for Sliding Cube-Shaped Robots in a Quadratic Number of Moves” *40th International Symposium on Computational Geometry*, 2024
9. “Super Guarding and Dark Rays in Art Galleries” *The Canadian Conference on Computational Geometry*, 2023
10. “Reconfiguration of 3D Pivoting Modular Robots” *SoCG Young Researchers Forum*, 2023
11. “Blockchain Ensured Physical Visitor Access Control and Authentication (poster)” *IEEE International Conference on Mobile Ad Hoc and Smart Systems, REUNS*, 2022

Accolades

UMass Lowell Student Research Symposium - Winner Kennedy College of Sciences (Graduate) 2023
Magna Cum Laude - Rochester Institute of Technology 2021